Name: _____

TOPIC TEST 1

Chapter 8: Arithmetic and geometric sequences

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 8 short answer questions (16 marks)
- Part C: 2 analysis questions (6 marks)

Part A

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 What is the common difference of the sequence 23, 32, 41, … ?

A 23

B 10

C 9

D $\frac{32}{23}$

E $\frac{23}{32}$

[1 mark]

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2 What is the common ratio of the sequence $-45, 15, -5, \ldots$?

A 45

B 30

C $\frac{1}{3}$

D -30

E $-\frac{1}{3}$

[1 mark]

3 What is the twentieth term of the sequence $3, 8, 13, \ldots$?

A 98

B 20

C 100

D 103

E 93

[1 mark]

4 What is the eighth term of the series $3 - 6 + 12 - \ldots$?

A 8

B 1536

C -1536

D 384

E -384

[1 mark]

5 Which of the following series has a limiting sum?

A $2 + 3 + 4 + \cdots$

B $-2 + 3 - 4 + \cdots$

C $4 + 3 + 2 + \cdots$

D $4 - 2 + 1 - \cdots$

E $4 - 3 + 2 - \cdots$

[1 mark]

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Part B

8 short answer questions
16 marks
Show your working where appropriate.

1 Evaluate to 7 terms: $4 + 7 + 10 + 13 + \cdots$

[2 marks]

2 Evaluate to 8 terms: $3 - 6 + 12 - 24 + \cdots$

[2 marks]

3 Find the limiting sum of $3 + 2 + \frac{4}{3} + \frac{8}{9} + \frac{16}{27} + \cdots$

[2 marks]

4 The first and last term of an AP containing 29 terms are 16 and 128 respectively. What is the sum of all terms in the sequence?

[2 marks]

5 The first and fifth terms of a GP are 5 and 3125 respectively. What is S_5 for this sequence?

[2 marks]

6 The last term of an arithmetic sequence is 642 and the common difference is 13. How many terms are positive?

[2 marks]

7 The fifth term of a geometric sequence is 243 and the seventh term is 2187. What is the sum of the first 6 terms?

[2 marks]

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8 Using your knowledge of a sum to infinity, write $0.6\overline{58}$ as a simplified fraction. Show all working.

[2 marks]

Part C

2 analysis questions
6 marks
Show your working where appropriate.

1 Bailey bought a one-bedroom unit for $360 000. He paid $90 000 cash and took out a loan for the remainder. Under the terms of the loan, Bailey agreed to repay $1500 each month plus interest of 6% p.a. on the unpaid balance. If he followed this agreement until the loan was paid out, how much did he pay altogether?

[3 marks]

2 Merenda bought a one-bedroom unit for $360 000. She paid $90 000 cash and took out a loan for the remainder. Under the terms of her loan, Merenda agreed to make a payment each month with interest of 6% p.a. on the unpaid balance until the loan was paid out. Her loan was taken out over 20 years, how much did she pay altogether?

[3 marks]
[Total marks: 27]

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Answers

Part A

[1 mark each]

1 Common difference $= 41 - 32 = 32 - 23 = 9$

$\therefore$ **C** [1 mark]

2 Common ratio $= \dfrac{-5}{15} = \dfrac{15}{-45} = -\dfrac{1}{3}$

$\therefore$ **E** [1 mark]

3 $t_{20} = 3 + 19 \times 5 = 98$

$\therefore$ **A** [1 mark]

4 $t_8 = 3 \times (-2)^7 = -384$

$\therefore$ **E** [1 mark]

5 $r = \dfrac{-2}{4} = \dfrac{1}{-2} = -\dfrac{1}{2}, \therefore |r| < 1$

$\therefore$ **D** [1 mark]

Part B

1 $4 + 7 + 10 + \cdots + 22$ [1 mark]

$= \dfrac{7}{2}(4 + 22) = 91$ [1 mark]

2 $3 - 6 + 12 - 24 + 48 - 96 + 192 - 384$ [1 mark]

$= \dfrac{(-2)(-384) - 3}{-2 - 1} = \dfrac{765}{-3} = -255$ [1 mark]

3 $3 + 2 + \dfrac{8}{9} + \dfrac{24}{27} + \cdots$ [1 mark]

$= \dfrac{3}{1 - \dfrac{2}{3}} = \dfrac{9}{3 - 2} = 9$ [1 mark]

4 $S_n = \dfrac{n}{2}(a + l)$

$S_{29} = \dfrac{29}{2}(16 + 128)$ [1 mark]

$S_{29} = 2088$ [1 mark]

5 $t_5 = 5 \times r^4 = 3125$

$\therefore r^4 = 625, r = 5$ [1 mark]

$S_5 = \dfrac{5(3125 - 1)}{5 - 1}$

$= \dfrac{15\,620}{4} = 3905$ [1 mark]

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6 $642 + 629 + 616 + \cdots$

$\therefore 642 + (n-1) \times (-13) > 0$ **[1 mark]**

$-13n > -655$

$n < 50\frac{5}{13}$

$n = 50$ **[1 mark]**

7 $ar^4 = 243$

$ar^6 = 2187$

$\therefore r^2 = 9$

$r = \pm 3$ **[1 mark]**

$S_6 = \dfrac{ar^6 - a}{r - 1}$

$= \dfrac{2187 - 3}{\pm 3 - 1} = 1092, -546$ **[1 mark]**

8 $0.6\overline{58} = 0.6 + \frac{58}{1000} + \frac{58}{100000} + \frac{58}{10000000} + \cdots$

$S_\infty = 0.6 + \dfrac{\frac{58}{1000}}{1 - \frac{1}{100}}$ **[1 mark]**

$= \dfrac{6}{10} + \dfrac{58}{1000 - 10}$

$= \dfrac{6 \times 99}{990} + \dfrac{58}{990}$

$= \dfrac{652}{990} = \dfrac{326}{495}$ **[1 mark]**

Part C

1 Cash = \$90 000, Loan = \$270 000

First payment = $\$1500 + 0.005 \times 270\,000 = \$2850.00$

Second payment = $\$1500 + 0.005 \times 268\,500 = \$2842.50$

Third payment = $\$1500 + 0.005 \times 267\,000 = \$2836.00$

Last payment = $\$1500 + 0.005 \times 1500 = \$1507.50$

From the first three, each payment decreases by \$7.50. **[1 mark]**

$S_{180} = \dfrac{180}{2}[2850.00 + 1507.50]$

$= \$392\,175$ **[1 mark]**

Total paid = $\$90\,000 + \$392\,175 = \$482\,175$ **[1 mark]**

2 $A_1 = \$270\,000 \times 1.005 - M$

$A_2 = \$270\,000 \times 1.005^2 - M \times 1.005 - M$

$A_3 = \$270\,000 \times 1.005^3 - M \times 1.005^2 - M \times 1.005 - M$

$\vdots$

$A_{240} = \$270\,000 \times 1.005^{240} - M \times 1.005^{239} - M \times 1.005^{238} - \cdots - M$ **[1 mark]**

Since $A_{240} = 0$, by rearranging this expression we get

$$M = \frac{\$270\,000 \times 1.005^{240}\left(1.005 - 1\right)}{\left(1.005^{240} - 1\right)} = \$1934.36$$ **[1 mark]**

$\therefore$ Total paid $= 240 \times \$1934.36 + \$90\,000$

$= \$554\,247.33$ **[1 mark]**

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